The Art of Prediction

Predicting Stock Prices: Mathematics, Algorithms and the Limits of Market Intelligence

Why stock price prediction matters?

In 1970, economist Eugene Fama introduced an idea that still unsettles traders today: stock prices already reflect everything the market knows. He called it the efficient market hypothesis, it implies that beating the market consistently is can prove to be incredibly difficult. And yet, people keep trying.

Stock prices touch nearly every corner of finance, from how pension funds are managed to how derivatives are priced. The pursuit to find a reliable signal in the noise has driven some of the most sophisticated technology ever built. Machine learning, in particular, has become the dominant toolkit for those who believe the market still has secrets to give up.

1900: The Random Walk
Louis Bachelier publishes The Theory of Speculation, proposing that stock prices move randomly.

1970: The EMH Framework
Eugene Fama formally defines the Efficient Market Hypothesis, claiming you cannot "beat the market."

1973: The Quant Revolution
The Black-Scholes Model is released, providing a mathematical formula for pricing derivatives.

1988: Renaissance Technologies
The Medallion Fund is launched, proving that sophisticated algorithms can find "signals" in the noise.

2002: Behavioral Finance
Daniel Kahneman wins the Nobel Prize, proving that human psychological bias creates predictable market "irrationality."

2010: The Flash Crash
A massive, sudden crash highlights the unintended consequences of high-speed algorithmic trading.

2020's: AI & Deep Learning
Prediction shifts to "Alternative Data," using AI to analyze news sentiment, satellite imagery and social trends.

The reality is more humbling than the hype. Markets absorb new information with startling speed, adjusting almost instantly to earnings reports, central bank decisions or political shocks. Day-to-day price movements are largely random and the patterns that do exist are fragile. This is precisely what makes the problem so compelling and so resistant to easy answers.

How algorithmic trading works?

A traditional algorithmic trading system follows a straightforward pipeline: collect data, apply a set of predefined rules and convert those rules into buy or sell orders. The logic is explicit, using conditions like moving average crossovers or threshold signals. If the market does X, then do Y. There is no learning, it's just execution.

Machine learning based systems act a little differently. Rather than being told what to look for, the model learns relationships between inputs and outputs from historical data, then applies the extrapolated patterns to the live market.

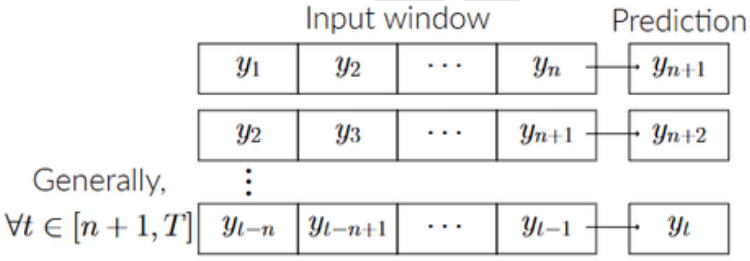
$$y = f(x)$$

Here, x might represent inputs such as price, volume or news, while y is the predicted signal or value. The function f represents the relationship the system has learned between past data and future outcomes in numerical form. The developer's role shifts from writing rules to curating data and shaping the learning process.

In practice, the historical data is divided into a training set and a test set. The model learns on the first portion, then is evaluated on unseen data to check whether its patterns hold up beyond the sample it was trained on.

For time-series problems, this learning process usually relies on rolling input windows, where past observations are used to predict the next value.

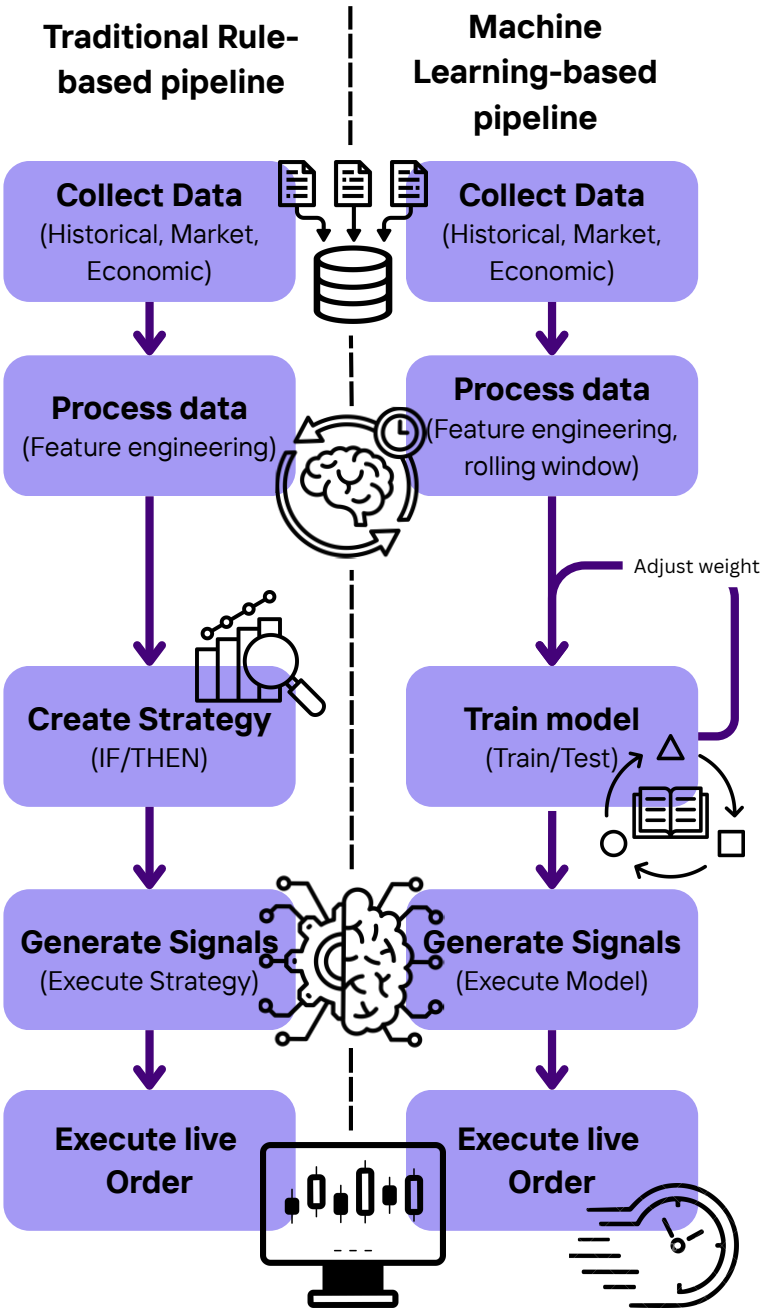
$$(y_{t-n}, \cdots, y_{t-1}) \rightarrow y_t$$



Neural networks push this further by learning more complex non-linear relationships from these inputs.

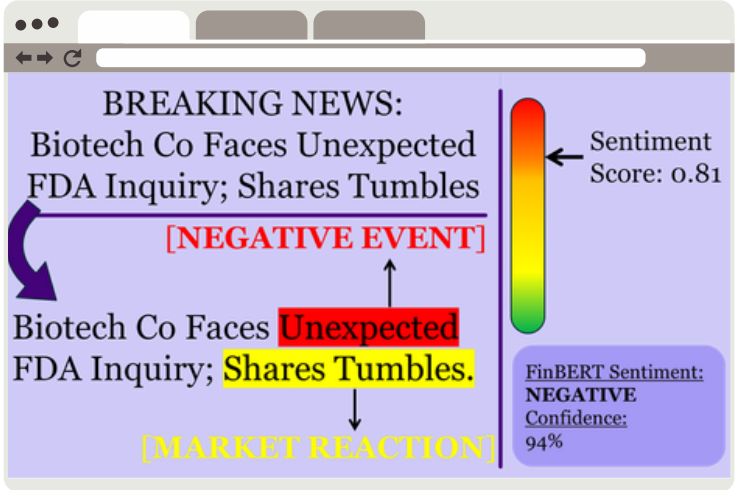
$$a = \sigma(Wx + b)$$

In simple terms, this equation is the process by which the network transforms inputs into patterns it can recognise, which makes them especially useful in noisy and complex markets.

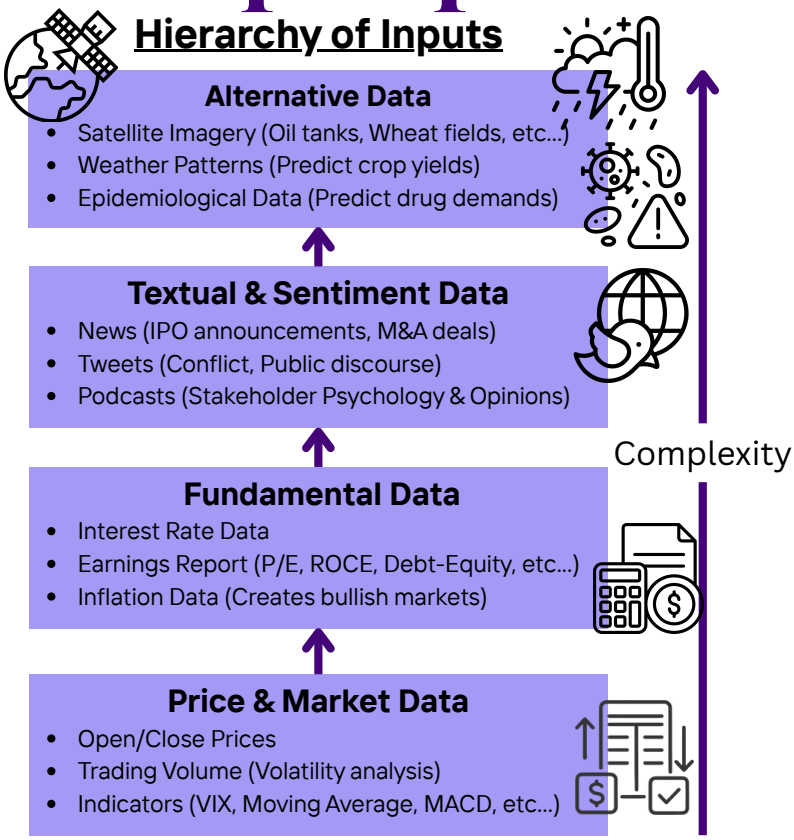


What can we use to help us predict stock prices?

There are many types of input available to a predictive model. Price-based features such as returns, rolling averages and momentum indicators are the most common starting point, capturing short-term pattern within the market. Fundamental data such as earnings reports and interest rate changes, operate on a slower timescale, offering a window into longer-term valuation rather than day-to-day noise.



Text has become an increasingly powerful input in modern markets. Transformer models, the "T" in ChatGPT, can interpret news headlines and corporate announcements with a nuanced understanding of tone and sentiment. For example, a model like FinBERT can analyse a headline about a corruption scandal and classify it as negative, enabling a trading system to reduce exposure before prices react. It might seem that combining more data types would lead to better market predictions, but more inputs create problems of their own. Not every variable carries meaningful information; many simply add noise.



Issues from inefficient data include:

- **Overfitting**: the model memorises patterns that only existed in past data.
- **Underfitting**: the model is too simple to capture meaningful structure.
- **Vanishing or exploding gradients**: in deep networks, training can become ineffective or numerically unstable.

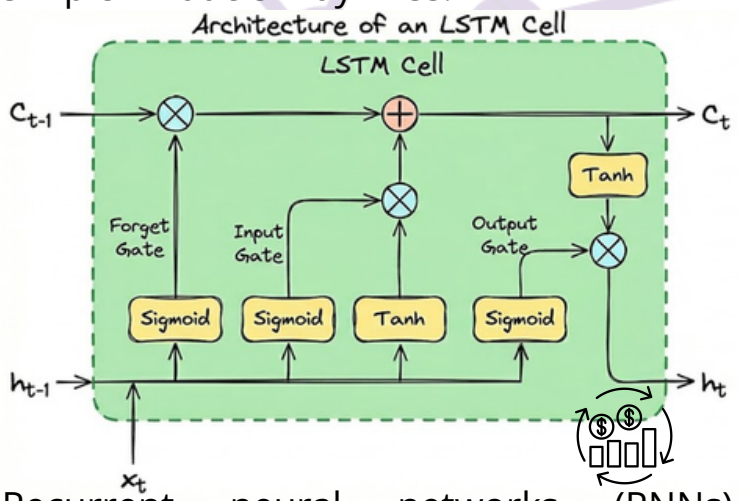
Practitioners try to manage these issues with techniques such as:

- **Dropout**
- **Early stopping**
- **Gradient clipping**

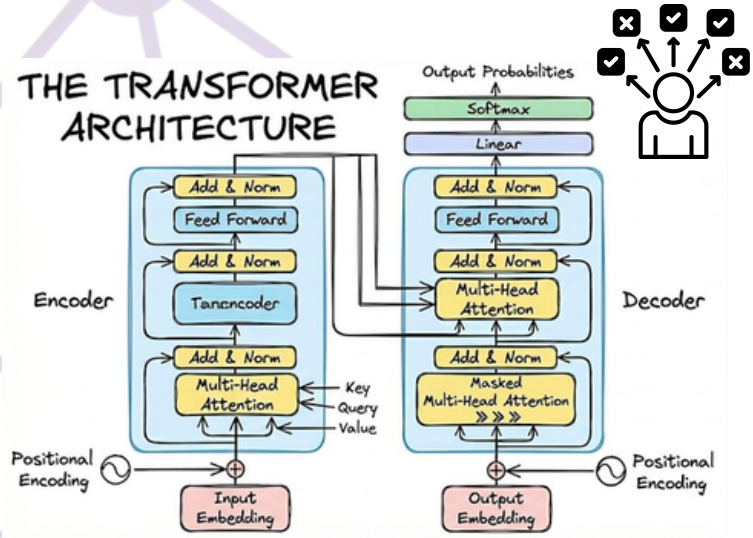
In fast-moving markets, a slightly less accurate model that responds in microseconds can outperform a more precise one that takes too long to decide. Model design is therefore both a statistical and an engineering problem. Spread Networks famously laid 827 miles of fibre optic cable between Chicago and New Jersey to cut latency by just a few milliseconds.

Which AI to use?

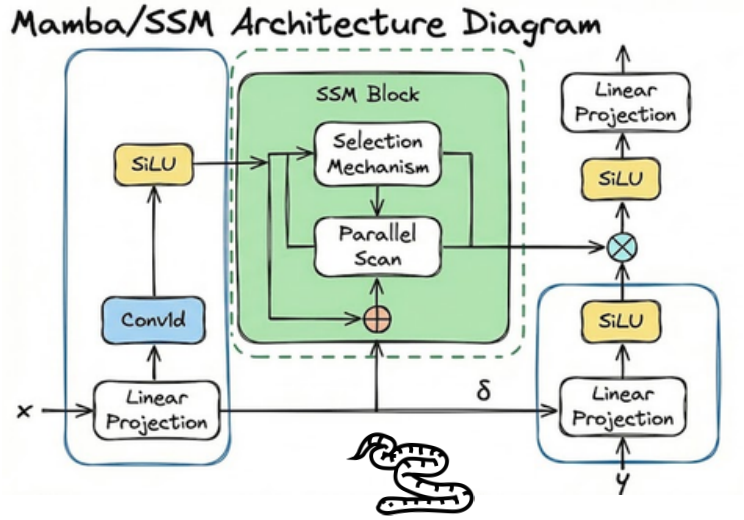
The choice of an artificial intelligence model depends strongly on the type and structure of the data being analysed. Linear models are often used as baselines because they are mathematically transparent and computationally efficient. These models can be effective when signals are weak and data is limited. Where relationships are more nonlinear or interactions between variables are more complex, machine learning models such as decision trees and neural networks can capture patterns that simpler models may miss.



Recurrent neural networks (RNNs), including long short-term memory (LSTM) models, are designed to process sequences one step at a time. This makes them well suited to time series data such as stock prices, where the order of observations matters. LSTMs use a gated architecture that helps mitigate the vanishing and exploding gradient problem, making them more effective than standard RNNs at retaining information across longer sequences. They can capture short- to medium-term dependencies, such as momentum or mean reversion patterns, but still often struggle with very long sequences or rapidly changing market conditions.



Transformers take an entirely different approach, processing entire sequences at once through attention mechanisms, which work by scoring how relevant each data point is to every other, allowing the model to focus on what matters most. In a trading context, this makes them particularly powerful for integrating different types of data together, with price history, macroeconomic indicators and news sentiment all handled within a single framework.



More recent architectures, such as Mamba models, are designed to handle long sequences efficiently while maintaining lower computational cost. They aim to combine the sequential strengths of RNNs with the scalability of transformers, making them well-suited to real-time trading systems where both speed and long-range memory are important.

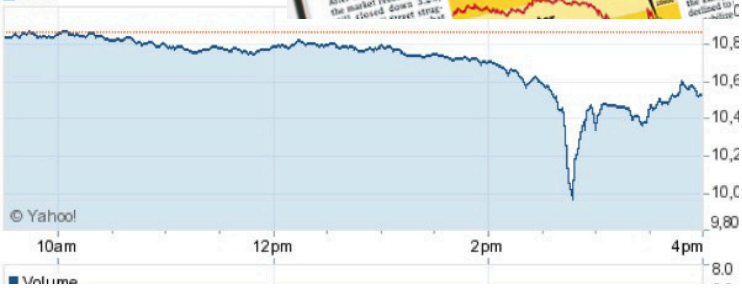
Ethics & Limits of prediction.

The ethical questions surrounding algorithmic trading are not simply about performance, but about what happens when thousands of automated systems respond simultaneously to similar signals. When models interact faster than any human can observe, volatility can be amplified in ways that are difficult to predict or control. These risks are not merely hypothetical. The trader Navinder Singh Sarao, the so-called "Hound of Hounslow", was linked to disruptive activity around the 2010 Flash Crash, placing and rapidly cancelling large orders in a way that exposed just how sensitive markets had become to automated behaviour. The crash wiped nearly a trillion dollars from US equity markets in minutes before partially recovering. Automated systems, it turned out, do not operate in isolation: they react to one another and that feedback can escalate instability with alarming speed.

THE WALL STREET JOURNAL.

Market Plunge Baffles Wall Street

Trading Glitch Suspected in 'Mayhem' as Dow Falls Nearly 1,000, Then Bounces



This complexity also makes modern trading systems harder to govern. Neural networks, in particular, rarely offer a clear explanation for a given decision, which raises serious questions about accountability when something goes wrong. Speed and data access further distribute advantage unevenly, giving well capitalised firms an edge that smaller participants cannot match.

Conclusion.

Taken together, these limits suggest that the real challenge is not whether markets can be predicted, but how narrow and temporary any predictive edge is likely to be. The efficient market hypothesis does not say prediction is impossible, only difficult. A model trained on years of data can still fail when inflation surprises, a central bank shifts tone or a geopolitical shock changes sentiment overnight. Markets are not governed by mathematics alone; they are shaped by competition and by human events. What makes prediction worth pursuing is the discipline it demands. Building systems for noisy, shifting data forces better thinking about uncertainty, more rigorous testing and a more honest sense of what models can and cannot do. That rigour, however imperfect the results, is what keeps the field moving forward.

REFERENCES